Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Simulate cross correlation and autocorrelation on discrete time signals.**	
Experiment No: 3	**Date:29/07/2026**	**Enrollment No:92400133079**

Theory:

Cross-Correlation and Autocorrelation

Cross-correlation and autocorrelation are mathematical techniques used to determine the **degree of similarity between signals**. These methods are commonly applied in areas such as **digital signal processing, image processing, communication systems, and pattern recognition**.

Cross-correlation is used to compare two different signals by shifting one signal relative to the other. At each time shift, the corresponding values of the two signals are multiplied and summed. The resulting correlation values indicate how closely the two signals match at different **time lags**.

Autocorrelation is used to compare a signal with a shifted version of **the same signal**. In other words, it determines how similar a signal is to itself when it is shifted by different time intervals. The resulting autocorrelation values show the relationship between the signal and its delayed versions.

Flowchart

The flowchart of the program can be organized into the following steps:

1. **Define the input signals** by creating two discrete-time signals for cross-correlation, or a single signal for autocorrelation.
2. **Calculate the correlation** between the two signals for cross-correlation, or calculate the correlation of the signal with itself for autocorrelation.
3. **Plot the obtained correlation values** to visualize the relationship between the signals at different time lags.
4. **Display the correlation graph** along with the corresponding lag values.
5. **Save the generated correlation result** if required.

Program

```
import numpy as np
import matplotlib.pyplot as plt
def cross_correlation(signal1, signal2):
    # Compute the cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr
def autocorrelation(signal):
    # Compute the autocorrelation
```

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```
    auto_corr = np.correlate(signal, signal, mode='full')
    return auto_corr
# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10])
# Compute the cross-correlation
cross_corr = cross_correlation(signal1, signal2)
# Compute the autocorrelation

auto_corr = autocorrelation(signal1)
# Plot the cross-correlation and autocorrelation signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.subplot(2, 1, 2)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.tight_layout()
plt.show()
# Save the cross-correlation or autocorrelation signals (optional)
# np.savetxt('cross_correlation.txt', cross_corr, delimiter=',')
# np.savetxt('autocorrelation.txt', auto_corr, delimiter=',')
```

Explanation:

- Import **NumPy** for numerical calculations and **Matplotlib** for plotting the signals.
- The cross_correlation() function finds the correlation between two signals using np.correlate() with full mode.
- The autocorrelation() function finds the similarity of a signal with itself by using the same signal as both inputs.

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- Define the required discrete-time signals and calculate their cross-correlation and autocorrelation.
- Use matplotlib.pyplot.stem() to display the obtained correlation signals.
- The results can optionally be saved as text files using np.savetxt().
- Different input signals can be assigned to signal1 and signal2 to observe different results.
- NumPy and Matplotlib must be installed before running the program.

OUTPUT

